

L7: Lesson Summary



This lesson introduced you to the concept of **community** in the context of networks.

We contrasted the problem of community detection to the “*cleaner*” problem of graph partitioning, emphasizing that in the former we do not know the number of communities – or even if the network has any communities in the first place.

A key concept in the community detection problem has been the modularity of a network. This metric allows us to compare different community assignments on the same network – and to examine whether a network has a strong community structure in the first place.

We also saw however that the modularity metric has a finite resolution and that we cannot detect communities of a smaller size than that resolution.

We also saw a number of algorithms for community detection, including divisive and agglomerative hierarchical clustering algorithms, greedy modularity maximization, and the Louvain algorithm. The literature in this area includes 100s of algorithms but the most commonly used are those we covered here.

We also discussed the concept of hierarchical modularity – and how it relates to the relation between the clustering coefficient and node degree.

In the subsequent lesson, we will cover more advanced topics of community detection – including overlapping communities and dynamic communities. We will also cover how to compare and characterize different community structures.